

SEVVAL ULUS MEMIS

Computer Engineer | Data Science Researcher | Health & Geospatial AI
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RESEARCH PROFILE

Computer engineer and M.S. Data Science researcher working across health data science, machine learning, remote sensing, and geospatial analysis. Research experience includes clinical machine learning with anonymized patient data, sensor-based field data collection, satellite-derived environmental analysis, global geospatial site characterization, interpretable machine learning, and computer vision for real-time movement assessment. Current interests include spatial determinants of health, spatial data science, GeoAI, AI-assisted geospatial data discovery, open GIS, and human-centered health data tools.

Spatial Health Data Science • Social Determinants of Health • GeoAI • Remote Sensing • Machine Learning • Geospatial Data Discovery

EDUCATION

M.S. in Data Science - Maryville University

St. Louis, Missouri

- GPA: 3.95/4.0

Expected May 2027

B.E. in Computer Engineering - Karabük University

Karabük, Türkiye

- ABET-accredited engineering program.

May 2025

RESEARCH EXPERIENCE

Undergraduate Thesis Researcher - Clinical Machine Learning

for Carpal Tunnel Syndrome

2024 - 2025

- Analyzed anonymized patient data collected at a public research hospital in collaboration with certified physiotherapists.
- Developed and evaluated Python-based machine-learning classification workflows for clinical prediction; current methodological revision emphasizes leakage-aware validation, reliability, and transparent evaluation.
- Performed data cleaning, model evaluation, and visual analytics to communicate patterns in clinical variables and prediction behavior.

Research Assistant - Remote Sensing Lab, Saint Louis

University

St. Louis, Missouri

Jan 2025 - Aug 2025

- Performed sensor-based field measurements and supported standardized acquisition of spatial, soil, crop, and environmental data.
- Contributed to organization, quality control, and analysis-ready preparation of remotely sensed and field-measurement data.

Principal Investigator - TÜBİTAK 2209-A National Research

Grant

2023 - 2024

- Led a nationally funded project on AI-based physiotherapy movement accuracy detection.
- Developed a Python/OpenCV computer-vision workflow for frame-by-frame human-motion tracking and real-time automated movement assessment.

Artificial Intelligence Researcher - SimurgAI Lab

Oct 2021 - Dec 2024

- Developed and supported machine-learning and computer-vision projects in Python, including data processing, model evaluation, visualization, and technical reporting.

SELECTED SPATIAL & DATA SCIENCE RESEARCH

- **Global seismic site characterization:** contributed conceptualization, data curation, and formal analysis for a study of 759 nuclear power plant sites using USGS Global Vs30 and facility-location data; evaluated geographic patterns and site-condition differences across countries and reactor types.
- **Satellite environmental analysis:** co-authored a MODIS-based study of land-surface-temperature changes in Northern Alaska permafrost regions, connecting satellite-derived environmental data with spatial change analysis.
- **Interpretable ML for transportation data:** contributed software, visualization, resources, and manuscript review to a large-scale LTPP study using 82,682 measurements from 603 pavement sections, with leakage-aware evaluation and SHAP-based interpretation.

SELECTED PUBLICATIONS

Memis, M. A., & Ulus Memis, S. U. (2027). Global seismic site characterization of nuclear power plant sites using USGS Vs30 data. Nuclear Engineering and Technology, 59, 104595. <https://doi.org/10.1016/j.net.2026.104595>

Role: Conceptualization; Data curation; Formal analysis. Geospatial site characterization using a global USGS raster dataset and facility-location data.

Memis, M. A., & Ulus Memis, S. U. (2026). Monitoring Land Surface Temperature (LST) Changes in Northern Alaska's Permafrost Regions Using MODIS Data. Journal of Engineering in Industrial Research, 7(3), 91-99.

Focus: satellite-derived environmental monitoring and spatial change analysis.

Memis, M. A., & Ulus Memis, S. U. (2026). Interpretable machine learning framework for resilient modulus estimation of unbound granular base layers: Insights from the LTPP database. Transportation Geotechnics, 64, 102292. <https://doi.org/10.1016/j.trgeo.2026.102292>

Role: Software; Visualization; Resources; Writing - review & editing. Large-scale, leakage-aware ML with SHAP interpretability.

Memis, M. A., & Ulus Memis, S. U. (2026). Resilience Framework for Critical Infrastructure: Artificial Intelligence Applications in Post-Disaster Ground Damage Assessment. Journal of Engineering in Industrial Research, 7(4), 195-211.

<https://doi.org/10.48309/jeires.2026.569173.1337>

Review focus: AI/deep learning, remote sensing, UAV/satellite data, and multisensor approaches for infrastructure assessment.

Memis, M. A., Keskin, I., Demir, S., & Ulus Memis, S. U. (2025). Machine learning-based prediction of permafrost degradation and its implications on geotechnical infrastructure: A comprehensive review. AI in Civil Engineering, 4(1), 28.

Focus: machine learning, environmental change, and infrastructure implications.

TECHNICAL SKILLS

- Programming & Data: Python, R, SQL, C/C++; pandas, NumPy, SciPy; data cleaning, statistical analysis, visualization, reproducible workflows.
- Machine Learning: scikit-learn, Random Forest, Gradient Boosting, XGBoost, neural networks, cross-validation, leakage-aware validation, SHAP interpretability, PCA, clustering.
- Geospatial / Remote Sensing: GIS, spatial datasets, raster-based geospatial analysis, MODIS-derived environmental data, remote-sensing workflows, UAV/hyperspectral/multispectral imagery, sensor-based field data collection.
- Health Data: anonymized clinical tabular data, feature analysis, classification, model evaluation, and visual analytics.
- Computer Vision: OpenCV, image processing, frame-based analysis, human-motion tracking, automated visual assessment.

RELEVANT COURSEWORK

Machine Learning | Deep Learning | Artificial Intelligence | Text Mining | Pattern Recognition | Data Mining | Image Processing | Mathematical Modeling | Data Structures | Database Systems | Object-Oriented Programming

LEADERSHIP & LANGUAGES

Google Developer Student Club (GDSC), Co-Leader (Aug 2023 - May 2025) - Organized technical workshops and student-led coding activities.

IEEE Student Chapter, Member (Aug 2020 - May 2024) - Participated in technical seminars and engineering activities.

Languages: Turkish (Native) | English (IELTS Academic 7.0)